



Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

Fatima Jinnah women university



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DBMS PROJECT

LIBRARY MANGEMENT SYSTEM

SOURCE CODE:

Login java:

```
package libraryms;
import java.awt.BorderLayout;
import java.awt.EventQueue;
import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.JLabel;
import javax.swing.SwingConstants;
import javax.swing.JTextField;
import javax.swing.JPasswordField;
import java.awt.FlowLayout;
import java.awt.Font;
import java.awt.Color;
import javax.swing.JButton;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;

public class login extends JFrame {

    private static final long serialVersionUID = 1L;
    private JPanel contentPane;
    private JTextField textField;
    private JPasswordField passwordField;

    public static void main(String[] args) {
        EventQueue.invokeLater(new Runnable() {
            public void run() {
                try {
                    login frame = new login();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        });
    }

    public login() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);
        contentPane = new JPanel();
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblNewLabel_1_1 = new JLabel("username");
```

```

lblNewLabel_1_1.setBounds(115, 59, 68, 17);
lblNewLabel_1_1.setFont(new Font("Tahoma", Font.BOLD, 14));
contentPane.add(lblNewLabel_1_1);

textField = new JTextField();
textField.setBounds(233, 58, 162, 20);
textField.setBackground(new Color(255, 255, 153));
contentPane.add(textField);
textField.setColumns(10);

JLabel lblNewLabel = new JLabel("password");
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 14));
lblNewLabel.setBounds(115, 106, 68, 14);
contentPane.add(lblNewLabel);

passwordField = new JPasswordField();
passwordField.setBackground(new Color(255, 255, 153));
passwordField.setBounds(233, 105, 162, 20);
contentPane.add(passwordField);

JButton btnNewButton = new JButton("login");
btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
btnNewButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        openLibraryDashboard();
    }
});
btnNewButton.setBounds(111, 151, 89, 23);
contentPane.add(btnNewButton);

JButton btnNewButton_1 = new JButton("close");
btnNewButton_1.setFont(new Font("Tahoma", Font.BOLD, 14));
btnNewButton_1.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        System.exit(0);
    }
});
btnNewButton_1.setBounds(233, 151, 89, 23);
contentPane.add(btnNewButton_1);
}

protected void openLibraryDashboard() {
    librarymanagementsystem dashboard = new librarymanagementsystem ();
    dashboard.setVisible(true);
    this.dispose();
}
}

```

LIBRARYMANGEMENTSYSTEM:

```
package libraryms;

import javax.swing.JButton;

import javax.swing.JFrame;

import javax.swing.JPanel;

import javax.swing.border.EmptyBorder;

import java.awt.EventQueue;

import java.awt.Font;

import java.awt.event.ActionEvent;

import java.awt.event.ActionListener;

public class librarymanagementsystem extends JFrame {

    private static final long serialVersionUID = 1L;

    private JPanel contentPane;

    /**
     * Launch the application.
     */
    public static void main(String[] args) {
        EventQueue.invokeLater(new Runnable() {
            public void run() {
                try {
                    librarymanagementsystem frame = new librarymanagementsystem();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        });
    }
}
```

```

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

});

}

```

```

/**

```

```

 * Create the frame.

```

```

 */

```

```

public librarymanagementsystem() {

    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    setBounds(100, 100, 450, 300);

    contentPane = new JPanel();

    contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));

    setContentPane(contentPane);

    contentPane.setLayout(null);


    JButton btnNewButton = new JButton("BOOKS");

    btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 12));

    btnNewButton.setBounds(22, 11, 89, 23);

    btnNewButton.addActionListener(new ActionListener() {

        public void actionPerformed(ActionEvent e) {

            // Open BooksFrame when "BOOKS" button is clicked

```

```
        BOOKS booksFrame = new BOOKS();

        booksFrame.setVisible(true);

    }

});

contentPane.add(btnNewButton);


JButton btnNewButton_1 = new JButton("PUBLISHER");

btnNewButton_1.setFont(new Font("Tahoma", Font.BOLD, 12));

btnNewButton_1.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        // Open PublisherFrame when "PUBLISHER" button is clicked

        PUBLISHER publisherFrame = new PUBLISHER();

        publisherFrame.setVisible(true);

    }

});

btnNewButton_1.setBounds(121, 11, 89, 23);

contentPane.add(btnNewButton_1);


JButton btnNewButton_2 = new JButton("BORROWBY");

btnNewButton_2.setFont(new Font("Tahoma", Font.BOLD, 11));

btnNewButton_2.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        // Open BorrowByFrame when "BORROWBY" button is clicked

        BORROWBBY borrowByFrame = new BORROWBBY();
```

```

        borrowByFrame.setVisible(true);

    }

});

btnNewButton_2.setBounds(223, 11, 89, 23);

contentPane.add(btnNewButton_2);


JButton btnNewButton_3 = new JButton("MEMBER");

btnNewButton_3.setFont(new Font("Tahoma", Font.BOLD, 12));

btnNewButton_3.addActionListener(new ActionListener() {

    public void actionPerformed(ActionEvent e) {

        // Open MemberFrame when "MEMBER" button is clicked

        MEMBER memberFrame = new MEMBER();

        memberFrame.setVisible(true);

    }

});

btnNewButton_3.setBounds(322, 11, 89, 23);

contentPane.add(btnNewButton_3);


JButton btnNewButton_4 = new JButton("log out");

btnNewButton_4.setFont(new Font("Tahoma", Font.BOLD, 14));

btnNewButton_4.setBounds(162, 89, 89, 23);

contentPane.add(btnNewButton_4);

}

}

```

BOOKS JAVA:

package libraryms;

```
import java.awt.EventQueue;
```

```
import javax.swing.JFrame; import  
javax.swing.JPanel; import  
javax.swing.border.EmptyBorder; import  
javax.swing.JLabel; import  
javax.swing.JTextField; import  
java.awt.Font; import  
javax.swing.JButton;
```

```
public class BOOKS extends JFrame {
```

```
    private static final long serialVersionUID =  
1L; private JPanel contentPane; private  
JTextField textField; private JTextField  
textField_1; private JTextField textField_2;  
private JTextField textField_3; private  
JTextField textField_4; private JTextField  
textField_5;
```

```
    /**
```

```
    * Launch the application.
```

```
    */
```

```
public static void main(String[] args) {
```

```
    EventQueue.invokeLater(new Runnable() {  
        public void run() {
```

```
            try {
```

```
                BOOKS frame = new BOOKS();
```

```
                frame.setVisible(true);
```

```
            } catch (Exception e) {
```

```
                e.printStackTrace();
```



```

        }
    }
});

}

/**
 * Create the frame.
 */
public BOOKS() {
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setBounds(100, 100, 450, 300);          contentPane = new
JPanel();          contentPane.setBorder(new EmptyBorder(5, 5,
5, 5));

        setContentPane(contentPane);
contentPane.setLayout(null);

        JLabel lblNewLabel = new JLabel("BOOK_ID:");
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 14));
        lblNewLabel.setBounds(62, 54, 91, 14);
        contentPane.add(lblNewLabel);

        JLabel lblNewLabel_1 = new JLabel("AUTHOR:");
lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 14));
        lblNewLabel_1.setBounds(62, 79, 91, 14);
        contentPane.add(lblNewLabel_1);

        JLabel lblNewLabel_2 = new JLabel("TITLE:");
lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 14));

```

```
lblNewLabel_2.setBounds(62, 104, 91, 14);  
contentPane.add(lblNewLabel_2);
```

```
        JLabel lblNewLabel_3 = new JLabel("PRICE:");  
lblNewLabel_3.setFont(new Font("Tahoma", Font.BOLD, 14));  
lblNewLabel_3.setBounds(62, 129, 91, 14);  
contentPane.add(lblNewLabel_3);
```

```
        JLabel lblNewLabel_4 = new JLabel("AVAILABLE:");  
lblNewLabel_4.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_4.setBounds(62, 154, 91, 14);  
contentPane.add(lblNewLabel_4);
```

```
        JLabel lblNewLabel_5 = new JLabel("PUBLISHER_ID:");  
        lblNewLabel_5.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_5.setBounds(62, 179, 91, 14);  
contentPane.add(lblNewLabel_5);
```

```
        textField = new JTextField();  
        textField.setBounds(192, 53, 214, 20);  
        contentPane.add(textField); textField.setColumns(10);
```

```
        textField_1 = new JTextField();  
        textField_1.setBounds(192, 78, 214, 20);  
contentPane.add(textField_1);        textField_1.setColumns(10);
```

```
        textField_2 = new JTextField();  
textField_2.setBounds(192, 103, 214, 20);  
contentPane.add(textField_2);        textField_2.setColumns(10);
```

```

        textField_3 = new JTextField();
textField_3.setBounds(192, 128, 214, 20);
contentPane.add(textField_3);           textField_3.setColumns(10);

        textField_4 = new JTextField();
textField_4.setBounds(192, 152, 214, 20);
contentPane.add(textField_4);           textField_4.setColumns(10);

        textField_5 = new JTextField();
textField_5.setBounds(192, 177, 214, 20);
contentPane.add(textField_5);
        textField_5.setColumns(10);

        JButton btnNewButton = new JButton("OK");
        btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
        btnNewButton.setBounds(126, 227, 89, 23);
        contentPane.add(btnNewButton);
    }
}

```

BORROWBBY:

```
package libraryms;
```

```
import java.awt.EventQueue;
```

```
import javax.swing.JFrame; import
javax.swing.JPanel; import
javax.swing.border.EmptyBorder; import
javax.swing.JLabel; import
javax.swing.JTextField; import

```

```
java.awt.Font; import
```

```
javax.swing.JButton;
```

```
public class BOOKS extends JFrame {
```

```
    private static final long serialVersionUID =  
1L; private JPanel contentPane; private  
JTextField textField; private JTextField  
textField_1;
```

```
    private JTextField textField_2;  
private JTextField textField_3; private  
JTextField textField_4; private  
JTextField textField_5;
```

```
/**
```

```
* Launch the application.
```

```
*/
```

```
public static void main(String[] args) {
```

```
    EventQueue.invokeLater(new Runnable() {
```

```
        public void run() {
```

```
            try {
```

```
                BOOKS frame = new BOOKS();
```

```
                frame.setVisible(true);
```

```
            } catch (Exception e) {
```

```
                e.printStackTrace();
```

```
            }
```

```
        }
```

```
    });
```

```
}
```

```

    /**
    * Create the frame.
    */
    public BOOKS() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);          contentPane = new
        JPanel();

        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));

        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblNewLabel = new JLabel("BOOK_ID:");
        lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 14));
        lblNewLabel.setBounds(62, 54, 91, 14);
        contentPane.add(lblNewLabel);

        JLabel lblNewLabel_1 = new JLabel("AUTHOR:");
        lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 14));
        lblNewLabel_1.setBounds(62, 79, 91, 14);
        contentPane.add(lblNewLabel_1);

        JLabel lblNewLabel_2 = new JLabel("TITLE:");
        lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 14));
        lblNewLabel_2.setBounds(62, 104, 91, 14);
        contentPane.add(lblNewLabel_2);

        JLabel lblNewLabel_3 = new JLabel("PRICE:");
        lblNewLabel_3.setFont(new Font("Tahoma", Font.BOLD, 14));

```

```
lblNewLabel_3.setBounds(62, 129, 91, 14);
```

```
contentPane.add(lblNewLabel_3);
```

```
JLabel lblNewLabel_4 = new JLabel("AVAILABLE:");
```

```
lblNewLabel_4.setFont(new Font("Tahoma", Font.BOLD, 12));
```

```
lblNewLabel_4.setBounds(62, 154, 91, 14);
```

```
contentPane.add(lblNewLabel_4);
```

```
JLabel lblNewLabel_5 = new JLabel("PUBLISHER_ID:");
```

```
lblNewLabel_5.setFont(new Font("Tahoma",
```

```
Font.BOLD, 12)); lblNewLabel_5.setBounds(62, 179, 91,
```

```
14);
```

```
contentPane.add(lblNewLabel_5);
```

```
textField = new JTextField();
```

```
textField.setBounds(192, 53, 214, 20);
```

```
contentPane.add(textField);
```

```
textField.setColumns(10);
```

```
textField_1 = new JTextField();
```

```
textField_1.setBounds(192, 78, 214, 20);
```

```
contentPane.add(textField_1); textField_1.setColumns(10);
```

```
textField_2 = new JTextField();
```

```
textField_2.setBounds(192, 103, 214, 20);
```

```
contentPane.add(textField_2); textField_2.setColumns(10);
```

```
textField_3 = new JTextField();
```

```
textField_3.setBounds(192, 128, 214, 20);
```

```
contentPane.add(textField_3); textField_3.setColumns(10);
```

```

        textField_4 = new JTextField();
        textField_4.setBounds(192, 152, 214, 20);
        contentPane.add(textField_4);
        textField_4.setColumns(10);

        textField_5 = new JTextField();
        textField_5.setBounds(192, 177, 214, 20);
        contentPane.add(textField_5);
        textField_5.setColumns(10);

        JButton btnNewButton = new JButton("OK");
        btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));
        btnNewButton.setBounds(126, 227, 89, 23);
        contentPane.add(btnNewButton);
    }

}

```

MEMBER JAVA:

```

package libraryms;

import java.awt.EventQueue;

import javax.swing.JFrame; import
javax.swing.JPanel; import
javax.swing.border.EmptyBorder; import
javax.swing.JLabel; import
javax.swing.JTextField; import
java.awt.Font; import
javax.swing.JButton;

```

```

public class MEMBER extends JFrame {
    private static final long serialVersionUID = 1L;
    private JPanel contentPane;
    private JTextField textField;
    private JTextField textField_1;
    private JTextField textField_2;    private
    JTextField textField_3;    private
    JTextField textField_4;

    /**
     * Launch the application.
     */
    public static void main(String[] args) {
        EventQueue.invokeLater(new Runnable() {
            public void run() {
                try {
                    MEMBER frame = new MEMBER();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        });
    }

    /**
     * Create the frame.
     */

```



```
public MEMBER() {  
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
    setBounds(100, 100, 450, 300);  
        contentPane = new JPanel();  
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));  
  
        setContentPane(contentPane);  
contentPane.setLayout(null);  
  
        JLabel lblNewLabel = new JLabel("MEMBER_ID:");  
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel.setBounds(44, 66, 86, 14);  
contentPane.add(lblNewLabel);  
  
        JLabel lblNewLabel_1 = new JLabel("NAME:");  
lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_1.setBounds(44, 91, 86, 14);  
contentPane.add(lblNewLabel_1);  
  
        JLabel lblNewLabel_2 = new JLabel("MEMEBR_DATE:");  
        lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_2.setBounds(44, 116, 86, 14);  
contentPane.add(lblNewLabel_2);  
  
        JLabel lblNewLabel_3 = new JLabel("EXPIRY_DATE:");  
lblNewLabel_3.setFont(new Font("Tahoma", Font.BOLD, 11));  
        lblNewLabel_3.setBounds(44, 141, 86, 14);  
        contentPane.add(lblNewLabel_3);  
}
```

```
JLabel lblNewLabel_4 = new JLabel("ADDRESS:");  
lblNewLabel_4.setFont(new Font("Tahoma", Font.BOLD, 11));  
lblNewLabel_4.setBounds(44, 166, 86, 14);  
contentPane.add(lblNewLabel_4);
```

```
textField = new JTextField();  
textField.setBounds(175, 63, 214, 20);  
contentPane.add(textField);  
textField.setColumns(10);
```

```
textField_1 = new JTextField();  
textField_1.setBounds(175, 88, 214, 20);  
contentPane.add(textField_1);          textField_1.setColumns(10);
```

```
textField_2 = new JTextField();  
textField_2.setBounds(175, 113, 214, 20);  
contentPane.add(textField_2);          textField_2.setColumns(10);
```

```
textField_3 = new JTextField();  
textField_3.setBounds(175, 138, 214, 20);  
contentPane.add(textField_3);          textField_3.setColumns(10);
```

```
textField_4 = new JTextField();  
textField_4.setBounds(175, 163, 214, 20);  
contentPane.add(textField_4); textField_4.setColumns(10);
```

```
JButton btnNewButton = new JButton("OK");  
btnNewButton.setFont(new Font("Tahoma", Font.BOLD, 14));  
btnNewButton.setBounds(120, 211, 89, 23);  
contentPane.add(btnNewButton);
```

}

}

PUBLISHER JAVA:

package libraryms;

import java.awt.EventQueue;

import javax.swing.JFrame; import

javax.swing.JPanel; import

javax.swing.border.EmptyBorder; import

javax.swing.JLabel; import

javax.swing.JTextField; import

java.awt.Font; import

javax.swing.JButton;

public class PUBLISHER extends JFrame {

private static final long serialVersionUID = 1L;

private JPanel contentPane;

private JTextField textField; private

JTextField textField_1; private

JTextField textField_2;

/**

*** Launch the application.**

***/**

public static void main(String[] args) {

EventQueue.invokeLater(new Runnable() {

public void run() {

```

        try {
            PUBLISHER frame = new PUBLISHER();
            frame.setVisible(true);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

});

}

/**
 * Create the frame.
 */
public PUBLISHER() {
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setBounds(100, 100, 450, 300);        contentPane = new
    JPanel();

        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));

        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblNewLabel = new JLabel("PUBLISHER_ID:");
        lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 12));
        lblNewLabel.setBounds(51, 74, 134, 14);
        contentPane.add(lblNewLabel);

        JLabel lblNewLabel_1 = new JLabel("NAME:");
        lblNewLabel_1.setFont(new Font("Tahoma", Font.BOLD, 12));

```

```
lblNewLabel_1.setBounds(51, 113, 79, 14);  
contentPane.add(lblNewLabel_1);
```

```
        JLabel lblNewLabel_2 = new JLabel("ADDRESS:");  
lblNewLabel_2.setFont(new Font("Tahoma", Font.BOLD, 12));  
lblNewLabel_2.setBounds(51, 147, 79, 14);  
contentPane.add(lblNewLabel_2);
```

```
        textField = new JTextField();  
textField.setBounds(199, 72, 186, 20);  
contentPane.add(textField);  
textField.setColumns(10);
```

```
        textField_1 = new JTextField();  
textField_1.setBounds(199, 111, 186, 20);  
contentPane.add(textField_1);  
        textField_1.setColumns(10);
```

```
        textField_2 = new JTextField();
```

```
textField_2.setBounds(199,  
145, 186, 20);  
contentPane.add(textField_2);  
textField_2.setColumns(10);
```

```
JButton btnNewButton = new  
JButton("OK");  
btnNewButton.setFont(new  
Font("Tahoma", Font.BOLD, 14));  
btnNewButton.setBounds(120, 193, 80,  
23); contentPane.add(btnNewButton)
```

}

}

DatabaseConnection:

package connection;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;

public class DataConnection {

private static final String URL =
"jdbc:oracle:thin:@localhost:1521:orcl"; //
Update with your database URL

private static final String USER = "hr"; //
Update with your database username

private static final String PASSWORD =
"oracle"; // Update with your database
password

private static Connection connection =
null;

public static Connection getConnection() {

if (connection == null) {

try {

// Load the JDBC driver

Class.forName("com.mysql.cj.jdbc.Driver");

// Establish the connection

connection =
DriverManager.getConnection(URL, USER,
PASSWORD);

```

        System.out.println("Connection
established successfully.");
    } catch (ClassNotFoundException |
SQLException e) {
        e.printStackTrace();
        System.out.println("Failed to
establish connection.");
    }
}
return connection;
}
}

```

MAIN:

```
package lms;
```

```
import connection.DataConnection;
```

```
import java.sql.Connection;
```

```
import java.sql.SQLException;
```

```
public class libms {
```

```
    public static void main(String[] args) {
```

```
        // Get a connection to the database
```

```
        Connection connection =
DataConnection.getConnection();
```

```
        if (connection != null) {
```

```
            try {
```

```
                // Use the connection for your
database operations
```



```
    // For example, you can create  
statements, execute queries, etc.
```

```
    // Don't forget to close the  
connection when done
```

```
        connection.close();
```

```
    } catch (SQLException e) {
```

```
        e.printStackTrace();
```

```
    }
```

```
}
```

```
}
```

```
}
```